United International University

Department of Computer Science & Engineering

Final Examination – Fall 2024

Program: MSCSE | Course: CSE 6213 – Natural Language Processing | Full Marks: 40

Answer ANY FOUR questions. Each question carries 10 marks.

Verified answers with complete workings are provided for all six questions.

QUESTION 1 [10 Marks]

Corpus of 8 eating sentences:

1. We eat rice
2. I eat rice
3. He eats rice
4. Rahim eats meat
5. I eat mango
6. They eat mango
7. I do not eat fish
8. He does not eat banana

(a) Regular Expression [2 marks]

The regex that covers all 8 sentences:

```
RE = ((We|I|They) eat | (He|Rahim) eats | (I do|He does) not eat)
(rice|mango|fish|meat|banana)
```

Breakdown of components:

Component	Matches
(We|I|They) eat	Sentences 1, 2, 5, 6 (1st/2nd/3rd plural + base verb)

(He|Rahim) eats	Sentences 3, 4 (3rd person singular + inflected verb)
I do not eat	Sentence 7 (1st person negative with auxiliary "do")
He does not eat	Sentence 8 (3rd singular negative with auxiliary "does")
(rice|mango|fish|meat|banana)	Object NP (all possible food items)

(b) Context-Free Grammar (CFG) [2 marks]

Production rules:

```
S       →  NP1 VP_pos                    # We/I/They eat ...

S       →  NP2 VP3s                      # He/Rahim eats ...

S       →  NP1 AUX NEG V NP_OBJ          # I do not eat ...

S       →  NP2 VP_NEG                    # He does not eat ...


NP1     →  We | I | They

NP2     →  He | Rahim

NP_OBJ  →  rice | mango | fish | meat | banana


VP_pos  →  eat NP_OBJ                    # present base form

VP3s    →  eats NP_OBJ                   # 3rd person singular

VP_NEG  →  AUXS NEG V NP_OBJ             # negated VP for He/Rahim


AUX     →  do

AUXS    →  does

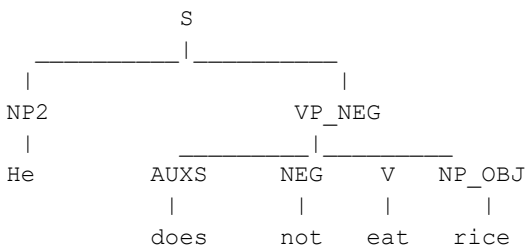
NEG     →  not

V       →  eat
```

(c) Parse Trees for "He does not eat rice" [2 marks]

Using rule: S → NP2 VP_NEG and VP_NEG → AUXS NEG V NP_OBJ

Top-Down Parse Tree:



Bottom-Up Parse Tree (Reductions):

```
Step 1:  He            → NP2
Step 2:  does          → AUXS
Step 3:  not           → NEG
Step 4:  eat           → V
Step 5:  rice          → NP_OBJ
Step 6:  AUXS NEG V NP_OBJ  → VP_NEG   [does not eat rice]
Step 7:  NP2 VP_NEG         → S         ACCEPT ✓
```

```
Tree (bottom-up view):
  S
  ├─ NP2  ──►  He
  └─ VP_NEG
     ├─ AUXS  ──►  does
     ├─ NEG   ──►  not
     ├─ V     ──►  eat
     └─ NP_OBJ ──► rice
```

(d) Leftmost and Rightmost Derivation for "He does not eat rice" [2 marks]

Leftmost Derivation (always expand leftmost non-terminal first):

```
S
  ⇒  NP2 VP_NEG                    [S → NP2 VP_NEG]
  ⇒  He VP_NEG                     [NP2 → He]
  ⇒  He AUXS NEG V NP_OBJ          [VP_NEG → AUXS NEG V NP_OBJ]
  ⇒  He does NEG V NP_OBJ          [AUXS → does]
  ⇒  He does not V NP_OBJ          [NEG → not]
  ⇒  He does not eat NP_OBJ        [V → eat]
  ⇒  He does not eat rice          [NP_OBJ → rice]   ✓
```

Rightmost Derivation (always expand rightmost non-terminal first):

```
S
  ⇒  NP2 VP_NEG                    [S → NP2 VP_NEG]
  ⇒  NP2 AUXS NEG V NP_OBJ         [VP_NEG → AUXS NEG V NP_OBJ]
  ⇒  NP2 AUXS NEG V rice           [NP_OBJ → rice]
```

```
⇒  NP2 AUXS NEG eat rice          [V → eat]
⇒  NP2 AUXS not eat rice          [NEG → not]
⇒  NP2 does not eat rice          [AUXS → does]
⇒  He does not eat rice           [NP2 → He]    ✓
```

(e) Shift-Reduce Parser for "He does not eat rice" [2 marks]

Grammar rule used: VP_NEG → AUXS NEG V NP_OBJ

Step	Input Buffer	Stack	Action
1	He does not eat rice $	$	SHIFT
2	does not eat rice $	$ He	REDUCE: He → NP2
3	does not eat rice $	$ NP2	SHIFT
4	not eat rice $	$ NP2 does	REDUCE: does → AUXS
5	not eat rice $	$ NP2 AUXS	SHIFT
6	eat rice $	$ NP2 AUXS not	REDUCE: not → NEG
7	eat rice $	$ NP2 AUXS NEG	SHIFT
8	rice $	$ NP2 AUXS NEG eat	REDUCE: eat → V
9	rice $	$ NP2 AUXS NEG V	SHIFT
10	$	$ NP2 AUXS NEG V rice	REDUCE: rice → NP_OBJ
11	$	$ NP2 AUXS NEG V NP_OBJ	REDUCE: AUXS NEG V NP_OBJ → VP_NEG
12	$	$ NP2 VP_NEG	REDUCE: NP2 VP_NEG → S
13	$	$ S	ACCEPT ✓

QUESTION 2 [10 Marks]

Corpus (6 sentences): We eat rice | I eat rice | He eats rice | Rahim eats rice | I eat mango | They eat mango

(a) Language Model Definition [1 mark]

A Language Model (LM) is a probability distribution over sequences of words. It assigns a probability $P(w_1, w_2, \ldots, w_n)$ to any word sequence, enabling a system to determine how likely a given sentence is in a language. LMs are the core of applications such as speech recognition, machine translation, auto-completion, and spell checking.

(b) Problems with Statistical Language Models [1 mark]

1. Data Sparsity (Zero Probability Problem):

Many valid word sequences never appear in the training corpus, so the model assigns them probability 0. For example, if "They eat rice" never appears in training, an n-gram model gives P("They eat rice") = 0, even though the sentence is perfectly grammatical.

2. Out-of-Vocabulary (OOV) Words:

Words not seen during training cannot be handled and break probability estimates for any sentence containing them.

Solution — Laplace (Add-1) Smoothing:

Add 1 to every count to avoid zero probabilities: P_smoothed(w) = (C(w) + 1) / (N + V) where V = vocabulary size.

(c) Unigram Probability of "They eat rice" [2 marks]

Step 1 — Count all words in corpus (6 sentences):

```
Words and frequencies:
  We(1), I(2), He(1), Rahim(1), They(1)          → 6 subject tokens
  eat(4), eats(2)                                  → 6 verb tokens
  rice(4), mango(2)                                → 6 object tokens
  Total tokens N = 18
```

Step 2 — Unigram counts:

Word	Count	P(word) = count / 18
They	1	1/18 ≈ 0.0556
eat	4	4/18 ≈ 0.2222
rice	4	4/18 ≈ 0.2222

Step 3 — Unigram LM assumes words are independent:

```
P(They eat rice) = P(They) × P(eat) × P(rice)
                 = (1/18) × (4/18) × (4/18)
                 = 16 / 5832
                 ≈ 0.00274
```

(d) Bigram and Trigram Probabilities of "We eat rice" [4 marks]

Bigram Model:

$P_bi(w_1 w_2 \ldots w_n) = P(w_1|\langle s\rangle) \times P(w_2|w_1) \times \ldots \times P(w_n|w_{n-1})$

Relevant bigram counts from corpus:

Bigram	Count	Unigram (context)	Bigram P
<s> We	1	C(<s>) = 6 sentences	P(We|<s>) = 1/6 ≈ 0.1667
We eat	1	C(We) = 1	P(eat|We) = 1/1 = 1.0000
eat rice	2	C(eat) = 4	P(rice|eat) = 2/4 = 0.5000

```
P_bi(We eat rice) = P(We|<s>) × P(eat|We) × P(rice|eat)
                  = (1/6) × 1.0 × 0.5
                  = 0.5/6
                  = 1/12  ≈  0.0833
```

Trigram Model:

$P_tri(w_1 w_2 \ldots w_n) = P(w_1|\langle s\rangle) \times P(w_2|\langle s\rangle, w_1) \times P(w_3|w_1, w_2) \times \ldots$

Relevant trigram counts from corpus:

Trigram	Count	Bigram (context)	Trigram P
<s> We eat	1	C(<s> We) = 1	P(eat|<s>,We) = 1/1 = 1.0
We eat rice	1	C(We eat) = 1	P(rice|We,eat) = 1/1 = 1.0

```
P_tri(We eat rice) = P(We|<s>) × P(eat|<s>,We) × P(rice|We,eat)
                   = (1/6) × 1.0 × 1.0
                   = 1/6  ≈  0.1667
```

Observation: Trigram probability (0.1667) > Bigram (0.0833) because the trigram captures the full context "We eat" which uniquely predicts "rice" in this corpus.

(e) Semantic Language Model [2 marks]

A Semantic Language Model extends the statistical n-gram model by incorporating word meanings and semantic relationships, rather than treating words as mere surface tokens.

Key characteristics:

Feature	Description
Representation	Words represented as dense vectors (embeddings) capturing

	semantic context
Training	Trained with neural networks (e.g., Word2Vec, GloVe, BERT) on large corpora
Similarity	Semantically similar words have similar vector representations
Context	Understands that "car" and "automobile" are interchangeable
Advantage over n-gram	Generalizes better; avoids the data sparsity problem by sharing probability mass among semantically similar words

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QUESTION 3 [10 Marks]

(a) Bag of Words for "I do not eat rice but eat fish" [4 marks]

Corpus documents:

Doc	Content
D1	I eat rice
D2	They eat rice
D3	He does not eat rice
D4	We do not eat fish
D5	I eat fish but not rice

Step 1 — Build vocabulary (union of all unique tokens, sorted):

```
V = { but,  do,  does,  eat,  fish,  he,  i,  not,  rice,  they,  we }

|V| = 11 terms
```

Step 2 — Tokenise test sentence: "I do not eat rice but eat fish"

```
Tokens: [i, do, not, eat, rice, but, eat, fish]  (8 words total)
```

Step 3 — Count Vector (raw term frequency):

but	do	does	eat	fish	he	i	not	rice	they	we
1	1	0	2	1	0	1	1	1	0	0

Step 4 — Frequency Vector (count ÷ total words = count ÷ 8):

but	do	does	eat	fish	he	i	not	rice	they	we
1/8	1/8	0	2/8	1/8	0	1/8	1/8	1/8	0	0

```
= [0.125, 0.125, 0, 0.25, 0.125, 0, 0.125, 0.125, 0.125, 0, 0]
```

(b) TF-IDF for the word "example" [3 marks]

Given documents:

Document	Content	Total Words
Doc1	This is a sample document	5
Doc2	This document is another example example example	7

TF-IDF formula: TF-IDF(t, d) = TF(t, d) × IDF(t)

where TF(t,d) = count(t in d) / total_words(d) and IDF(t) = $\log_{10}(N / df(t))$

Term	TF(example, Doc1)	TF(example, Doc2)	df	IDF	TF-IDF(Doc1)	TF-IDF(Doc2)
example	0/5 = 0	3/7 ≈ 0.4286	1	$\log_{10}(2/1)$ ≈ 0.3010	0 × 0.3010 = 0	0.4286 × 0.3010 ≈ 0.1290

Interpretation: "example" has TF-IDF = 0 in Doc1 (absent) and ≈ 0.1290 in Doc2 (distinctive).

(c) Hash Trick [3 marks]

Hash function: h(word) = ASCII_sum(word) mod table_size = ASCII_sum(word) mod 4

Step 1 — Compute hash values:

```
"RAT" : R=82, A=65, T=84  →  Sum = 231  →  231 mod 4 = 3   (bucket 3)
"MAT" : M=77, A=65, T=84  →  Sum = 226  →  226 mod 4 = 2   (bucket 2)
```

Step 2 — Hash Table (size 4):

Bucket Index	0	1	2	3
Words mapped	—	—	MAT	RAT

Step 3 — Feature Vector for document containing "RAT RAT MAT":

```
Bucket 0: 0  (no words mapped here)
Bucket 1: 0  (no words mapped here)
Bucket 2: 1  (MAT appears once)
Bucket 3: 2  (RAT appears twice)
```

```
Feature vector = [0, 0, 1, 2]
```

Note: Hash collisions can occur (different words mapping to the same bucket), but the trade-off is fixed low dimensionality regardless of vocabulary size.

QUESTION 4 [10 Marks] – Bangla Morphology

(a) Affixation in Bangla [2 marks]

Affixation is the morphological process of adding an affix (prefix, infix, or suffix) to a root/stem to create a new word or grammatical form.

i. Prefix (Upasarga – উপসর্গ): attached BEFORE the root

Prefix	Root	Derived Word	Meaning
অ-	শান্তি	অশান্তি	Unrest (negation prefix)
উপ-	দেশ	উপদেশ	Advice
নির-	জন	নির্জন	Deserted (without people)

ii. Infix (Causative infix -আন-): inserted WITHIN the root

Root Verb	Infix	Derived Verb	Meaning
পড়া	-আন-	পড়ানো	To cause to read / to teach
করা	-আন-	করানো	To cause to do / to get done
দেখা	-আন-	দেখানো	To cause to see / to show

iii. Suffix (Pratyay – প্রত্যয়): attached AFTER the root

Root	Suffix	Derived Word	Meaning
লেখ	-ক	লেখক	Writer (agentive noun)
সুন্দর	-ত্ব	সুন্দরত্ব	Beauty (abstract noun)
মানুষ	-গুলো	মানুষগুলো	The people (definite plural)

(b) Inflectional vs. Derivational Morphemes [2 marks]

Inflectional morphemes modify a word's grammatical form WITHOUT changing its category or core meaning. Derivational morphemes create NEW words, often changing the word category.

Type	Bangla Example	Morpheme	Effect
Inflectional (Tense)	খাওয়া → খাচ্ছি	-ছি	Verb stays verb; present progressive tense added
Inflectional (Number)	বই → বইগুলো	-গুলো	Noun stays noun; definite plural added
Inflectional (Case)	সে → তাকে	-কে	Pronoun stays pronoun; accusative case marked
Derivational (V→N)	লেখা → লেখক	-ক	Verb "write" → Noun "writer" (category changes)
Derivational (Adj→N)	সুন্দর → সুন্দরত্ব	-ত্ব	Adjective "beautiful" → Noun "beauty"
Derivational (V→N)	পড়া+শোনা → পড়াশোনা	compounding	Two verbs → one compound noun "study"

(c) Root and Stem [2 marks]

Root (মূল): the irreducible core form of a word, stripped of all affixes.

Stem: the base form to which inflectional affixes attach (may already contain derivational affixes).

Root examples (Dhatu – ধাতু):

Root (Dhatu)	Derived Forms
কর (do/make)	করা, করছি, করব, করেছি, করে (various tense/aspect forms)
লেখ (write)	লেখা, লেখক, লেখনী, লিখছি (various forms and derivatives)
যা (go)	যাওয়া, যাচ্ছি, যাবে, গেছি (go → went form change)

Stem examples:

Stem	Inflectional affixes attach as:	Example
লেখক (writer)	-গণ (formal plural)	লেখকগণ = "the writers" (formal)
ভালো (good)	-ত্ব, -বাসা	ভালোত্ব = "goodness"; ভালোবাসা = "love"
মানুষ (human)	-কে, -গুলো, -ত্ব	মানুষকে, মানুষগুলো, মানবত্ব

(d) Free and Bound Morphemes [2 marks]

Free morpheme: can stand alone as an independent word.

Bound morpheme: must attach to another morpheme; cannot stand alone.

Type	Bangla Example	Romanisation	Explanation
Free	বই	boi	"book" – independent word, complete meaning
Free	মাঠ	math	"field" – independent word
Free	ভালো	bhalo	"good" – independent adjective
Bound	-গুলো	-gulo	Plural/definite marker; must attach: বইগুলো
Bound	-কে	-ke	Accusative case; must attach: মানুষকে
Bound	-ছি	-chhi	Progressive suffix; must attach: খাচ্ছি = খা+ছি

(e) Clipping, Coining, and Acronyms [2 marks]

These are word-formation processes that create new vocabulary items.

i. Clipping — shortening an existing longer word:

Original (Bangla)	Clipped Form	Meaning
বিশ্ববিদ্যালয় (bishwobidyalay)	ভার্সিটি (varsity)	University (borrowed + clipped)
টেলিফোন (telephone)	ফোন (phone)	Phone
মোটরগাড়ি (motorgari)	মোটর (motor)	Motor car

ii. Coining — creating entirely new words for new concepts:

Coined Word	Origin	Meaning
কম্পিউটার	Latin "computare" (to calculate)	Computer
ইন্টারনেট	English technical coinage	Internet
মোবাইল	English "mobile" (shortened)	Mobile phone

iii. Acronyms — formed from initial letters of phrase:

Acronym	Full Form (Bangla)	English Meaning
র‍্যাব (RAB)	র‍্যাপিড অ্যাকশন ব্যাটালিয়ন	Rapid Action Battalion
বিজিবি (BGB)	বর্ডার গার্ড বাংলাদেশ	Border Guard Bangladesh
ঢাবি (DU)	ঢাকা বিশ্ববিদ্যালয়	Dhaka University

QUESTION 5 [10 Marks] – Phonetics and POS Tagging

(a) IPA Transcription of Five English Words [2 marks]

Word	IPA Transcription	Key phonetic note
guarantee	/ˌgær.ən.ˈtiː/	Primary stress on last syllable; schwa in middle

		syllable
ringtone	/ˈrɪŋ.toʊn/	Velar nasal /ŋ/; primary stress on first syllable
education	/ˌɛdʒ.ʊ.ˈkeɪ.ʃən/	Affricate /dʒ/; stress on 3rd syllable; final schwa
dogs	/dɒgz/	Voiced plural /z/ (after voiced /g/); British RP /ɒ/
history	/ˈhɪs.tər.i/	Primary stress on first syllable; reduced vowel in middle

(b) Orthographic vs. Phonetic Representation [2 marks]

Orthographic representation is the conventional written spelling of a word following standard rules.

Phonetic (phonemic) representation uses IPA symbols to show the actual sounds produced when the word is spoken.

Key difference: Written letters do not always match spoken sounds (silent letters, digraphs, etc.).

Word	Orthographic	Phonetic (IPA)	Discrepancy
knight	k-n-i-g-h-t	/naɪt/	"k", "g", "h" are silent; "igh" = /aɪ/
phone	p-h-o-n-e	/foʊn/	"ph" = /f/; final "e" is silent
through	t-h-r-o-u-g-h	/θruː/	"gh" silent; "ou" = /uː/; "th" = /θ/

(c) POS Tagger Confusion Matrix [1 mark]

4×4 confusion matrix (rows = Actual class, columns = Predicted class):

Actual \ Predicted	N (Noun)	V (Verb)	Adj	Adv	Row Total (Support)
N (Noun)	50	5	3	2	60
V (Verb)	4	45	3	3	55
Adj	2	2	40	6	50
Adv	3	3	3	41	50
Col Total	59	55	49	52	215

Diagonal cells (bold) = True Positives (correctly classified instances).

(d) Precision, Recall, and F1-Score per Class [3 marks]

Formulas:

```
Precision(class) = TP / (TP + FP)  =  TP / Column Total
Recall   (class) = TP / (TP + FN)  =  TP / Row Total (Support)
F1       (class) = 2 × Precision × Recall / (Precision + Recall)
```

Calculations:

Class	TP	Support (Row)	Col Total	Precision (TP/Col)	Recall (TP/Row)	F1-Score
N (Noun)	50	60	59	50/59 = 0.8475	50/60 = 0.8333	0.8403
V (Verb)	45	55	55	45/55 = 0.8182	45/55 = 0.8182	0.8182
Adj	40	50	49	40/49 = 0.8163	40/50 = 0.8000	0.8081
Adv	41	50	52	41/52 = 0.7885	41/50 = 0.8200	0.8040

Sample F1 verification for Noun:

```
F1(N) = 2 × 0.8475 × 0.8333 / (0.8475 + 0.8333)
      = 2 × 0.7063 / 1.6808
      = 1.4126 / 1.6808
      ≈ 0.8403  ✓
```

(e) Weighted Averages [2 marks]

Weighted average = Σ (support × metric) / total samples (215)

```
Weighted Precision = (60×0.8475 + 55×0.8182 + 50×0.8163 + 50×0.7885) / 215
                   = (50.850 + 45.001 + 40.815 + 39.425) / 215
                   = 176.091 / 215
                   ≈ 0.8190

Weighted Recall    = (60×0.8333 + 55×0.8182 + 50×0.8000 + 50×0.8200) / 215
                   = (50.000 + 45.000 + 40.000 + 41.000) / 215
                   = 176.000 / 215
                   ≈ 0.8186

Weighted F1-Score  = (60×0.8403 + 55×0.8182 + 50×0.8081 + 50×0.8040) / 215
                   = (50.418 + 45.001 + 40.405 + 40.200) / 215
                   = 176.024 / 215
                   ≈ 0.8187
```

Metric	Weighted Average
Precision	0.8190
Recall	0.8186
F1-Score	0.8187

QUESTION 6 [10 Marks] – Machine Translation & Similarity

(a) Transfer Approach for Bangla→English MT [2 marks]

Sentence: "বালকেরা মাঠে ক্রিকেট খেলিতেছে" (Boys are playing cricket in the field)

The Transfer Approach has three stages:

Stage 1 — Morphological/Syntactic Analysis (Source language):

Bangla Token	POS / Analysis	Morpheme breakdown
বালকেরা	NP (Subject)	বালক + -এরা (boy + plural marker)
মাঠে	PP (Locative)	মাঠ + -এ (field + locative case)
ক্রিকেট	NP (Object)	ক্রিকেট (loanword, no inflection)
খেলিতেছে	VP (Predicate)	খেল + -ইতেছে (play + present continuous)

Stage 2 — Transfer (Bilingual mapping using Transfer Lexicon):

Bangla Analysis	English Transfer	Notes
NP[বালকেরা]	NP[Boys]	Plural noun — no article in Bangla
PP[মাঠে]	PP[in the field]	Locative → "in the" + noun
NP[ক্রিকেট]	NP[cricket]	Direct loanword transfer
VP[খেলিতেছে, pres-cont]	VP[are playing]	Present continuous → be+V-ing

Stage 3 — Generation (Target language English):

Apply English syntactic rules (SVO order, add articles):

```
Generated:  Boys  are playing  cricket  in the field.
            ↑NP   ↑VP          ↑NP      ↑PP
```

(b) Corpus-Based (Example-Based) MT for "They are playing cricket in the paddy field" [3 marks]

In example-based MT, the system finds similar sentences in a parallel corpus and recombines translation fragments.

Suppose the parallel corpus contains:

#	English Source	Bangla Translation
1	Boys are playing cricket in the field	বালকেরা মাঠে ক্রিকেট খেলিতেছে
2	Farmers are working in the paddy field	কৃষকেরা ধান ক্ষেতে কাজ করিতেছে
3	They are eating in the kitchen	তারা রান্নাঘরে খাইতেছে

4	He is playing cricket	সে ক্রিকেট খেলিতেছে

Fragment matching for input: "They are playing cricket in the paddy field"

Input Fragment	Closest Corpus Match	Bangla Equivalent
"They are"	Corpus #3: "They are eating"	তারা
"playing cricket"	Corpus #1,#4: "playing cricket"	ক্রিকেট খেলিতেছে
"in the paddy field"	Corpus #2: "in the paddy field"	ধান ক্ষেতে

Combined translation (fragments recombined and reordered per Bangla syntax):

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তারা  ধান ক্ষেতে  ক্রিকেট  খেলিতেছে।
(They)  (paddy field-LOC)  (cricket)  (are playing)
= "They are playing cricket in the paddy field"  ✓
```

(c) Cosine and Jaccard Similarity [3 marks]

Documents after removing stop words (is, our, the, their, for, from, in, a):

Document	Original	After Stop-word Removal	Term Counts
Doc-I	"Our spoken language is Bangla. Many people sacrifice their lives for the language."	spoken, language, Bangla, many, people, sacrifice, lives, language	language=2, others=1
Doc-II	"Bangla is our language from heart. In language movement many people participate."	Bangla, language, heart, language, movement, many, people, participate	language=2, others=1

Vocabulary (union): spoken, language, Bangla, many, people, sacrifice, lives, heart, movement, participate

Term-frequency vectors:

	spoken	language	Bangla	many	people	sacrifice	lives	heart	movement	participate
v_1 (Doc-I)	1	2	1	1	1	1	1	0	0	0
v_2 (Doc-II)	0	2	1	1	1	0	0	1	1	1

Cosine Similarity:

$$v_1 \cdot v_2 = (1\times0)+(2\times2)+(1\times1)+(1\times1)+(1\times1)+(1\times0)+(1\times0)+(0\times1)+(0\times1)+(0\times1)$$
$$= 0 + 4 + 1 + 1 + 1 + 0 + 0 + 0 + 0 + 0 = 7$$

$$|v_1| = \sqrt{(1^2+2^2+1^2+1^2+1^2+1^2+0+0+0)} = \sqrt{(1+4+1+1+1+1+1)} = \sqrt{10} \approx 3.162$$
$$|v_2| = \sqrt{(0+4+1+1+1+0+0+1+1+1)} = \sqrt{10} \approx 3.162$$

$$\text{Cosine(Doc-I, Doc-II)} = 7 / (3.162 \times 3.162) = 7 / 10 = 0.70$$

Jaccard Similarity (set-based, binary):

Set(Doc-I) = { spoken, language, Bangla, many, people, sacrifice, lives } → 7 unique terms
Set(Doc-II) = { Bangla, language, heart, movement, many, people, participate } → 7 unique terms

Intersection = { language, Bangla, many, people } → |I| = 4
Union = { spoken, language, Bangla, many, people, sacrifice, lives, heart, movement, participate } → |U| = 10

Jaccard(Doc-I, Doc-II) = |Intersection| / |Union| = 4 / 10 = 0.40

Similarity Measure	Value	Interpretation
Cosine Similarity	0.70	High similarity — vectors point in similar directions (weighted)
Jaccard Similarity	0.40	Moderate overlap — 4 out of 10 unique terms shared

(d) IPA Transcription of Bangla Words [2 marks]

Bangla Word	Romanisation	IPA Transcription	Meaning
ক্রয়	kroy	/krɔj/	Purchase
দোকান	dokan	/doˈkan/	Shop / Store
ফুটবল	futbol	/ˈfʊt.bal/	Football
মেরামত	meramat	/me.ˈra.mot/	Repair

IPA symbols used: /ɔ/ = open-mid back unrounded vowel (ও-sound in ক্রয়); /ˈ/ = primary stress marker; /ʊ/ = near-close back rounded vowel.

— End of Answer Sheet —